

Yash Thube

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Research & applied interests

Generalist embodied agents that learn representations of spatial, temporal, physics dynamics, and interact with the physical world.

Broadly - Model based RL (representation learning, temporal abstraction, long horizon planning, robot learning), 3D Vision (scene understanding, spatial and causal reasoning), Architectures (VLMs, VLAs, World models, Diffusion).

Tools - PyTorch, OpenCV, HuggingFace (Transformers, TRL, Diffusers, PEFT), Git, NumPy, TorchVision, TransformerLens, Panda3D, Gym, PyBullet, MuJoCo, Pillow, Matplotlib.

Experience

- **AI Research Engineer**, Future Sportler – Remote 01/2026 – Present
Working on computer vision for sports analytics.
- **Machine Learning Engineer**, Hudl India – Pune, MH 03/2024 – 11/2024
Implemented YOLOv8 based pipeline for real time player detection and tracking in sports videos. Fine-tuned VLMs on custom sports datasets for semantic video understanding. Achieved 92% precision in crowded scenes using DeepSORT multi-object tracking.
- **Deep Learning Collaborator**, Omdena – remote, 10/2023 – 01/2024
Architected transformer-based pipeline for mental health sentiment analysis on social platforms. Implemented RAG with fine-tuned LLMs and developed audio classification models using CNNs for emotional tone detection in voice content.
- **Computer Vision Collaborator**, AI Accelerator Institute – remote, 02/2023 – 06/2023
Worked on segmentation and self supervised learning for vision.
- **Technical Writer**, InPlainEnglish | Towards AI – remote 04/2022 – 05/2023
Distilled complex technologies into clear and accessible content with primary focus on ML and AWS.

Projects & Research

- **MATS (arXiv preprint)** – A behavioral audit toolkit to detect pathological truth bias in VLMs, experiments include activation patching to causally localize failures in cross-attention layers and pooled representations across LLaVA, CLIP, and Qwen-VL architectures.
- **Paper Implementations** – PyTorch concept implementations of research papers, aimed at deepening my understanding of the underlying concepts.
- **Multimodal/VLMs Research Hub** – A technical resource for researchers exploring Vision-Language Models (VLMs) and multimodal learning, featuring seminal papers/models, datasets, benchmarks, ethical challenges, and research directions.

Github

Education

Savitribai Phule Pune University (SPPU)
B.E. Computer Engineering - 2026